HANDSMEN THREADS

SALESFORCE PROJECT DOCUMENTATION

Custom CRM, Data Model, Automation, Security and Testing

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Technology: Salesforce Platform Developer

Platform: Salesforce Developer Edition

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Document Control

Item	Details
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Important accuracy note: This report distinguishes configured items, intended behavior, and runtime results. A Setup screenshot proves that a configuration item exists or is active; it does not by itself prove that a flow executed successfully, an email was delivered, a validation rule blocked a save, or a user had the expected access. Record test outcomes only after executing the tests.

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Executive Summary

HandsMen Threads is a Salesforce-based CRM concept for a men's fashion retail business. It organizes customer, product, order, inventory and marketing-campaign information in custom objects. The project documentation covers the data model, validation controls, Flow Builder automation, email templates, user access, testing, deployment and maintenance.

The configuration evidence supplied for review identifies five principal custom objects: HandsMen Customer, HandsMen Product, HandsMen Order, Inventory and Marketing Campaign. It also describes active validation rules for order amount, inventory quantity and customer email; a Stock Status formula field; Stock Alert and Loyalty Status Update flows; email templates; scheduled processing entries; and a role hierarchy for Sales, Inventory and Marketing.

This document is deliberately evidence-based. Where the exact formula, trigger condition, recipient, schedule, Apex implementation or test result is not established by the supplied material, the report labels it as a verification item rather than presenting it as completed functionality.

1. Project Overview

1.1 Background

Retail teams need a reliable way to maintain customer details, track orders, monitor inventory and coordinate marketing communications. Disconnected spreadsheets or inconsistent manual updates can lead to duplicate data, missed stock replenishment and incomplete customer follow-up.

HandsMen Threads uses Salesforce as a shared workspace for these activities. Custom objects represent the core business records, validation rules are intended to reduce invalid data, and automation can notify responsible users or update customer loyalty information when configured criteria are met.

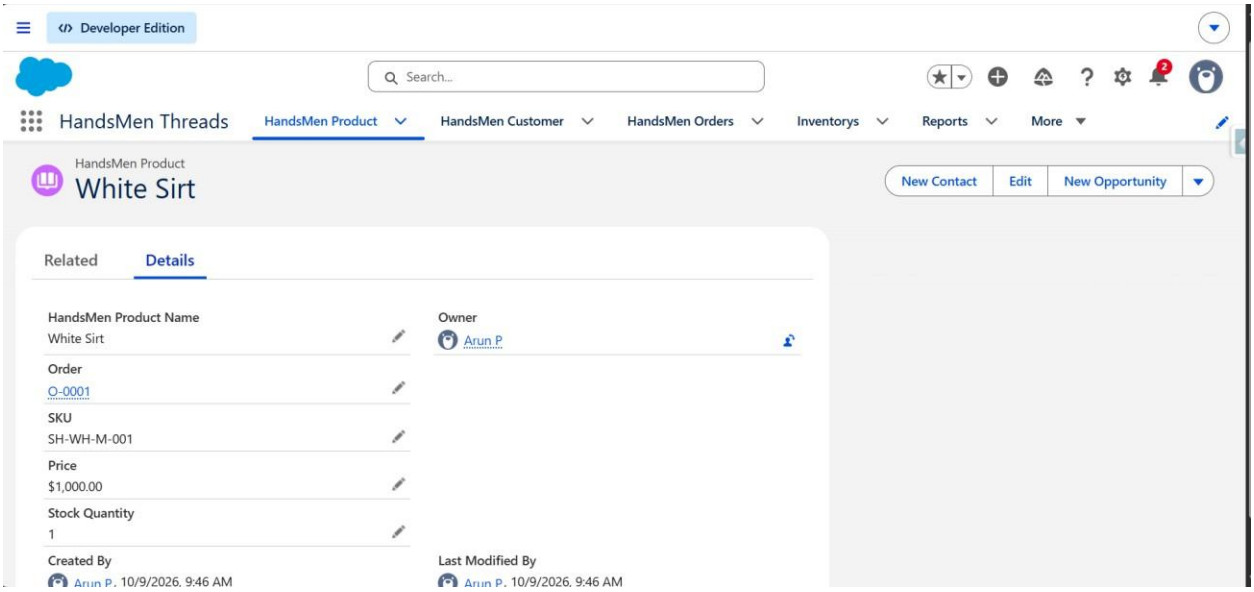
1.2 Business Problem

- Customer and purchase information needs to be stored consistently and linked through relationships.
- Product and stock information needs to be easy to review by operational users.
- Invalid quantities, amounts and email values should be prevented at record save time.
- Low-stock events and loyalty changes should be handled consistently rather than relying only on manual checks.
- Access to customer information and operational records should match each user's responsibilities.

1.3 Proposed Solution

The proposed solution uses Salesforce Lightning Experience, custom objects and fields, lookup/master-detail relationships, validation rules, formula fields, Flow Builder, email templates

and role/profile-based access. Apex, batch processing, reports, dashboards and other advanced features should be documented as implemented only when their metadata, source code or execution evidence is available.



2. Objectives and Scope

2.1 Objectives

- Centralize customer, product, order, inventory and marketing-campaign records.
- Connect related records through appropriate Salesforce relationships.
- Improve data quality using validation rules and calculated fields.
- Support stock monitoring and loyalty-status processing through automation.
- Define access responsibilities for Sales, Inventory, Marketing and administrators.
- Prepare repeatable test cases and retain evidence of expected and actual results.

2.2 In Scope

- Custom object and field configuration.
- Lookup and master-detail relationships.
- Validation rules and formula fields.
- Record-triggered and scheduled flows, where present in the org.
- Email templates and email-alert configuration, where present.
- Profiles, roles, permission sets, sharing and user access review.
- Functional testing, data-loading approach, deployment and maintenance planning.

2.3 Out of Scope / Not Assumed

This draft does not assume that Apex triggers, Apex classes, Batch Apex, Lightning Web Components (LWC), approval processes, duplicate rules, field history tracking, reports, dashboards or a CI/CD pipeline have been implemented. Add these as completed components only after confirming their existence and attaching suitable evidence.

3. Phase 1 — Requirement Analysis and Planning

3.1 Stakeholders and User Needs

Stakeholder	Need	Expected business value
Sales representative	Create and maintain customer and order records.	More reliable customer follow-up and order visibility.
Inventory manager	Maintain inventory quantities and warehouse details.	Earlier identification of stock shortages.
Marketing user	Associate campaign records with customer records and loyalty information.	More organized customer communication.
System administrator	Manage data quality, automation, permissions and monitoring.	More consistent configuration and support.

3.2 User Stories

- As a sales representative, I want customer and order records linked so that purchase activity can be followed up reliably.
- As an inventory manager, I want stock quantity and warehouse information available so that shortages can be identified.
- As a marketing user, I want campaigns associated with customers so that approved outreach can be planned.
- As an administrator, I want validation and access controls so that records remain consistent and visible only to authorized users.

3.3 Functional Requirements

ID	Requirement	Salesforce component
FR-01	Maintain customer contact details and loyalty information.	HandsMen Customer
FR-02	Maintain product catalogue details such as SKU and price.	HandsMen Product
FR-03	Record order customer, quantity, status and total amount.	HandsMen Order
FR-04	Maintain stock quantity and warehouse details linked to a product.	Inventory
FR-05	Record campaign start/end dates and associated customer.	Marketing Campaign
FR-06	Reject values that violate configured business constraints.	Validation Rules
FR-07	Notify the responsible user when configured low-stock conditions are met.	Stock Alert Flow / email alert, if active
FR-08	Evaluate customer loyalty status using the configured criteria.	Loyalty Status Update Flow, if active

3.4 Non-Functional Requirements

- Data consistency: required fields, field types and validation rules should match business rules.
- Security: object, field and record access should follow least-privilege principles.
- Maintainability: automation should have clear names, fault handling and documented owners.
- Traceability: each claimed feature should be supported by configuration evidence and, where applicable, runtime test evidence.
- Scalability: automation should avoid assumptions that only one record will be processed at a time.

3.5 Scope and Assumption Register

Item	Current documentation status	Action before final submission
Stock threshold	Exact threshold not verified from the supplied text.	Insert the actual formula/flow condition and screenshot.
Loyalty tiers	Gold/Silver/Bronze outcomes are described in the supplied report; full criteria need confirmation.	Verify thresholds, fields, default path and schedule.
Email delivery	Templates are described; delivery result is not established.	Run an approved test and capture evidence.
Apex / Batch Apex	Scheduled Apex entry is mentioned, but source code and class behavior are not fully documented.	Attach class, scheduler/batch details and test class only if implemented.
User access	Role names are shown, but role assignments and permissions need validation.	Test with representative users and capture results.

4. Phase 2 — Salesforce Environment and Data Model

4.1 Environment and Development Workflow

The project is documented for Salesforce Developer Edition using Lightning Experience. The recommended workflow is to configure and test changes in a development environment, record metadata and dependencies, validate changes in a test environment when available, and deploy only reviewed components to the target org.

- Record the org type, project owner and the date of each configuration change.
- Keep a change log for objects, fields, validation rules, flows, email templates and access settings.
- Do not include passwords, access tokens, private customer data or other secrets in screenshots or the report.

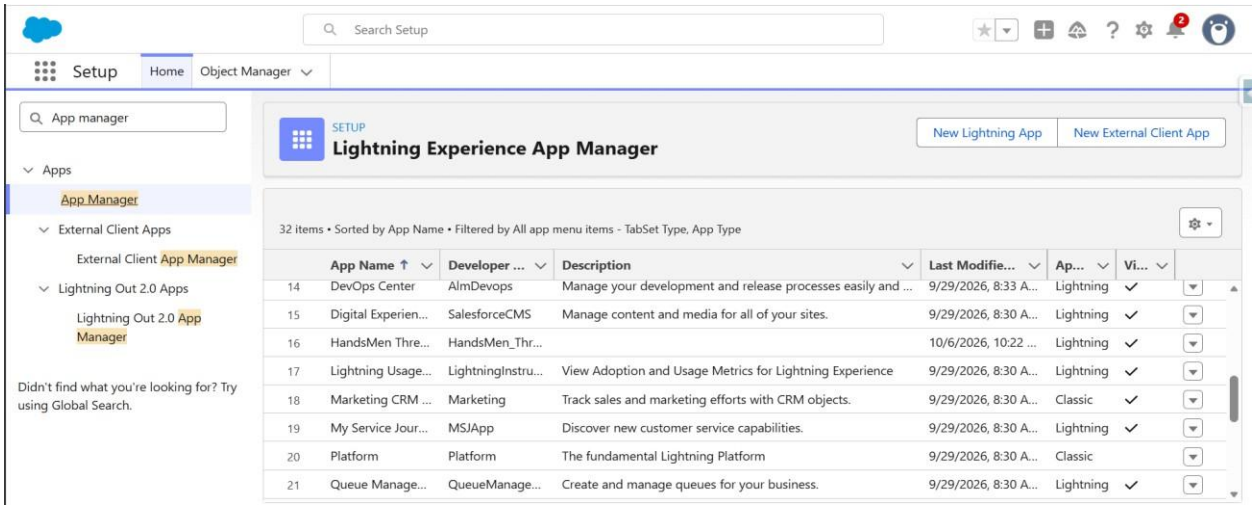


Figure 2. Salesforce org and App Manager .

4.2 Custom Objects

Object label	Purpose	Key documented fields / notes
HandsMen Customer	Stores customer contact and loyalty information.	Email, FirstName, LastName, Full Name formula, Loyalty Status, Phone, Total Purchases; confirm exact API names in Object Manager.
HandsMen Product	Stores product catalogue information.	Name (configured as text in the supplied report), Order lookup, Price__c, SKU__c, Stock_Quantity__c.
HandsMen Order	Stores customer purchase records.	Name configured as auto-number; Customer_c lookup, Quantity_c, Status__c, Total_Amount__c.
Inventory	Stores stock and warehouse information associated with a product.	Name configured as auto-number; Product_c master-detail, Stock_Quantity_c, Warehouse_c, Stock_Status__c formula.
Marketing Campaign	Stores campaign dates and associated customer.	Auto-number Name, Start_Date_c, End_Date_c, HandsMen_Customer__c lookup.

Field labels and API names above reflect the supplied project report and must be cross-checked against the live org before final submission. Salesforce custom fields typically use the __c suffix; standard fields may not.

4.3 Relationship Design

Parent / source object	Field	Relationship type shown	Meaning / design note
HandsMen Customer → HandsMen Order	HandsMen_Order.Customer__c	Lookup(HandsMen Customer)	An order can reference a customer.
HandsMen Order → HandsMen Product	HandsMen_Product.Order__c	Lookup(HandsMen Order)	The product record can reference an order; this direction should be reviewed against the desired business model.
HandsMen Product → Inventory	Inventory.Product__c	Master-Detail(HandsMen Product)	Inventory is a detail record under a product.
HandsMen Customer → Marketing Campaign	Marketing_Campaign.HandsMen_Customer__c	Lookup(HandsMen Customer)	A campaign record can reference a customer.

Data-model improvement: if one order can contain multiple products and one product can appear in many orders, a dedicated Order Item junction object is usually a better model. It can store the order, product, quantity, unit price and line total for each order line. This is a proposed enhancement, not a claim that the object already exists.

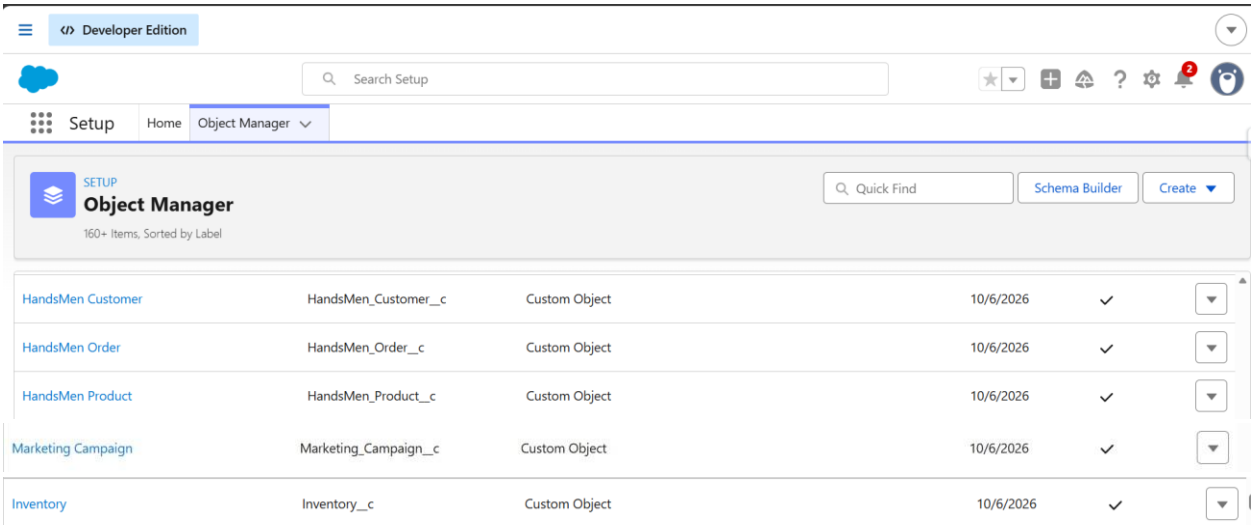


Figure 3. Object Manager overview showing the five custom objects.

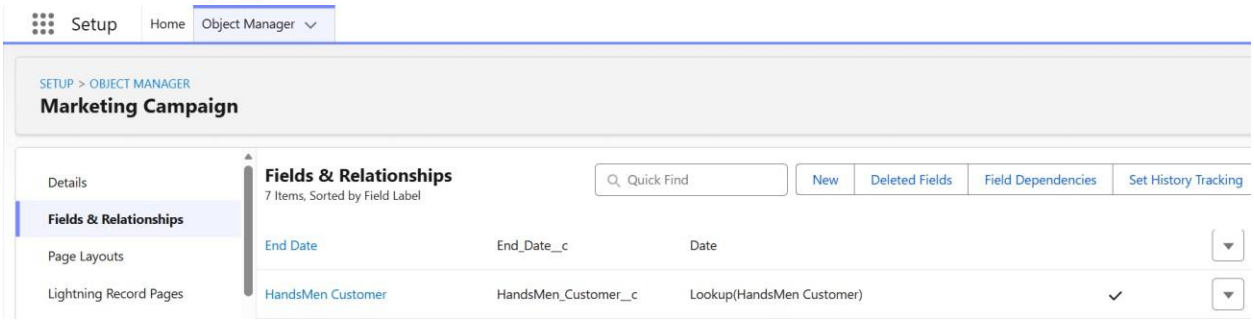


Figure 4. Relationship fields / schema evidence from Object Manager.

5. Object and Field Configuration

5.1 HandsMen Customer

HandsMen Customer is the customer master record used by orders and marketing campaigns. The supplied report identifies contact fields and loyalty-related fields, including Email, FirstName, LastName, Full Name, Loyalty Status, Phone and Total Purchases. Confirm each field's data type, required setting, formula expression and field-level security in Object Manager.

Field label (reported)	Type / configuration	Purpose	Verification needed
Email	Email or configured text Field	Customer contact and validation.	Confirm exact field type and Email validation formula.
FirstName / LastName	Text fields	Customer name components.	Confirm API names and required settings.
Full Name	Formula (if confirmed)	Readable full name.	Confirm formula and output.
Loyalty Status	Picklist (if confirmed)	Stores loyalty tier.	Confirm allowed values and default.
Phone	Phone	Customer contact.	Confirm field-level access.
Total Purchases	Number/currency/formula as configured	Supports loyalty criteria.	Confirm definition and whether it is maintained automatically.

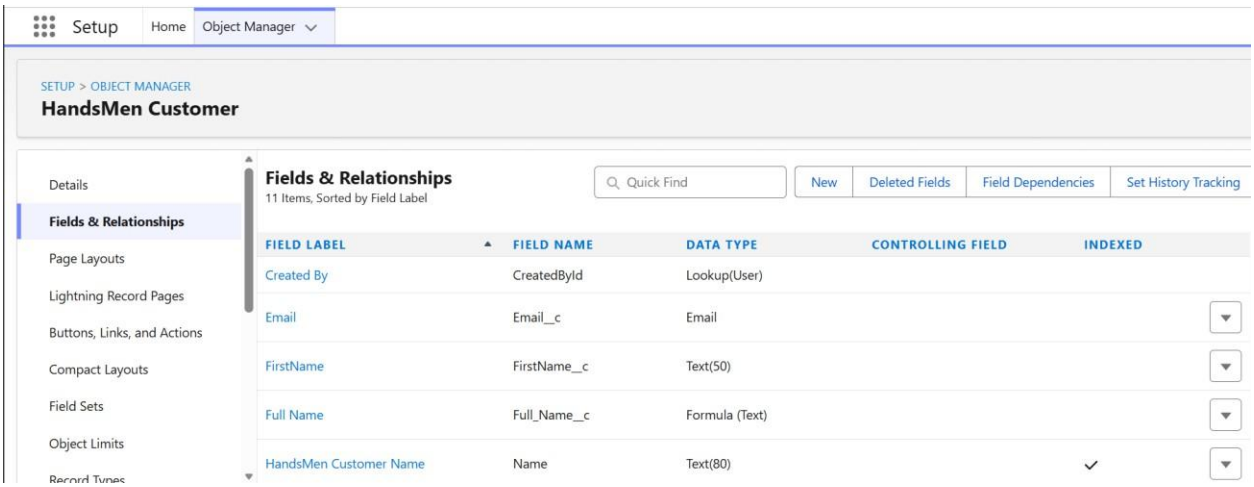


Figure 5. HandsMen Customer field list and field details.

5.2 HandsMen Product

HandsMen Product represents the catalogue. The supplied report lists a product name, SKU, price, stock quantity and an Order lookup. Confirm whether stock is maintained on the product, the Inventory object, or both; duplicate stock fields should have a clear source of truth to avoid inconsistent values.

Field label	Reported API name	Reported type	Purpose / check
HandsMen Product Name	Name	Text(80)	Product display name; confirm Name field settings.
Order	Order__c	Lookup(HandsMen Order)	Review relationship direction and business need.
Price	Price__c	Currency(18,0)	Confirm currency precision and whether decimals are needed.
SKU	SKU__c	Text(20)	Consider uniqueness if the business requires one SKU per product.
Stock Quantity	Stock_Quantity__c	Number(18,0)	Confirm whether this is a summary or a second stock source.

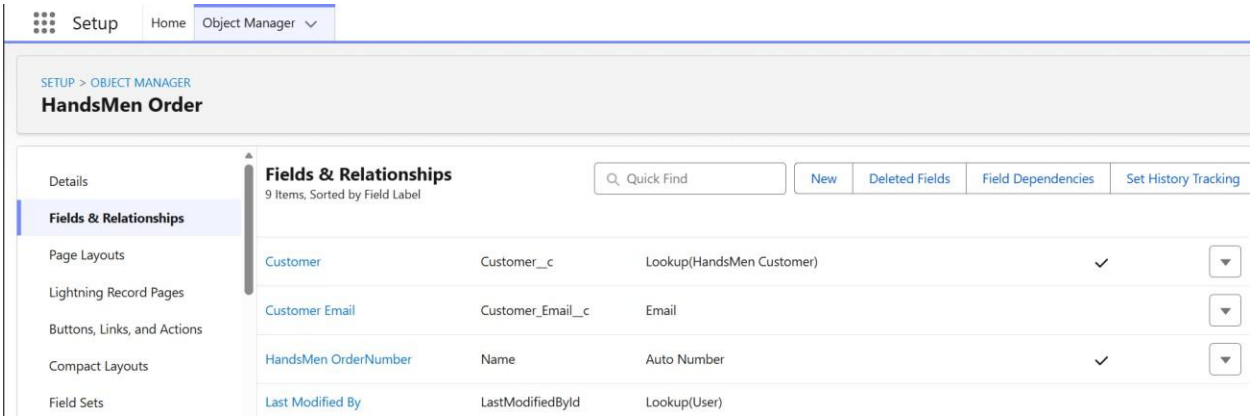


Figure 6. HandsMen Product fields and relationships.

5.3 HandsMen Order

Field label	Reported API name	Reported type	Purpose / check
HandsMen OrderNumber	Name	Auto Number	Unique order identifier generated by Salesforce.
Customer	Customer__c	Lookup(HandsMen Customer)	Links the order to a customer.
Quantity	Quantity__c	Number(18,0)	Confirm minimum and whether order items are required.
Status	Status__c	Picklist	Document actual values and transition rules.
Total Amount	Total_Amount__c	Number(18,0)	Confirm currency vs number type and amount calculation.

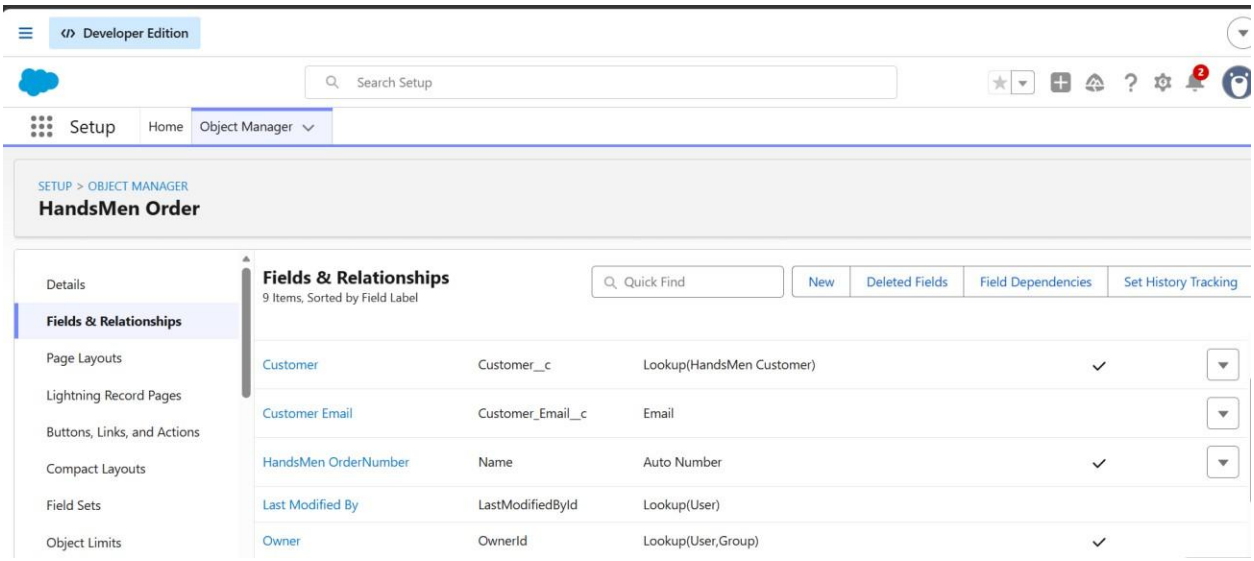


Figure 7. HandsMen Order fields and customer lookup.

5.4 Inventory

Field label	Reported API name	Reported type	Purpose / check
Inventory Name	Name	Auto Number	Inventory record identifier.
Product	Product__c	Master-Detail(HandsMen Product)	Required parent product relationship.
Stock Quantity	Stock_Quantity__c	Number(18,0)	Quantity on hand; negative values should be rejected if this is the rule.
Warehouse	Warehouse__c	Text(80)	Warehouse / storage location.
Stock Status	Stock_Status__c	Formula (Text)	Derived stock status; insert exact formula and threshold only after confirming.

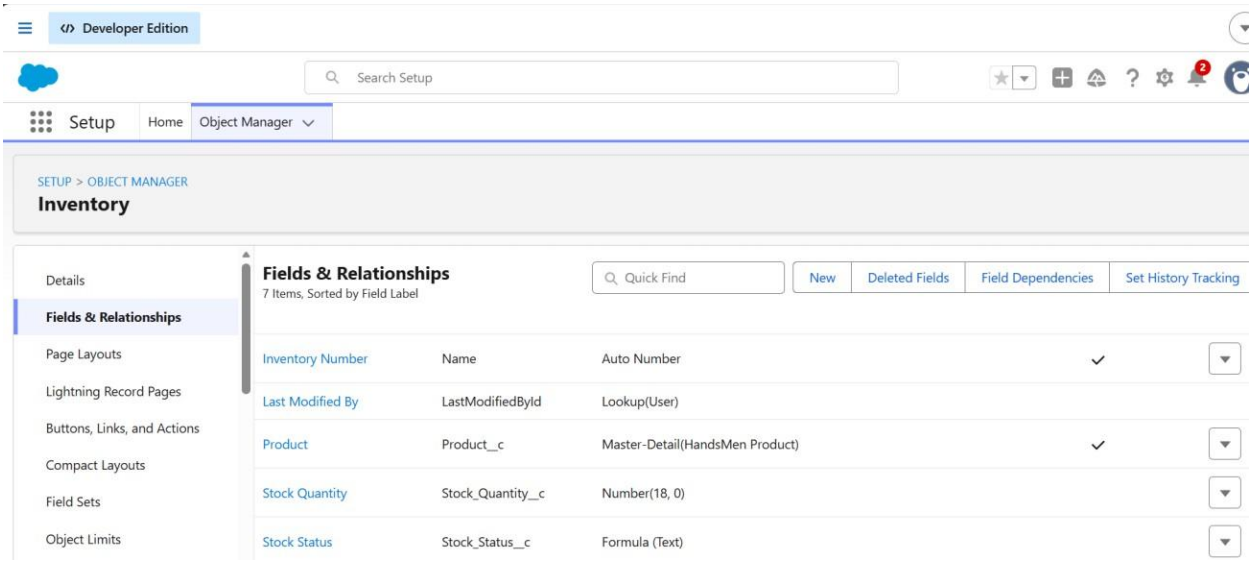


Figure 8. Inventory fields, relationship and Stock Status formula field.

5.5 Marketing Campaign

Field label	Reported API name	Reported type	Purpose / check
Marketing Campaign Name	Name	Auto Number	Campaign identifier.
Start Date	Start_Date__c	Date	Campaign start date.
End Date	End_Date__c	Date	Campaign end date; consider validation that end date is not before start date.
HandsMen Customer	HandsMen_Customer__c	Lookup(HandsMen Customer)	Associates the campaign record with a customer.

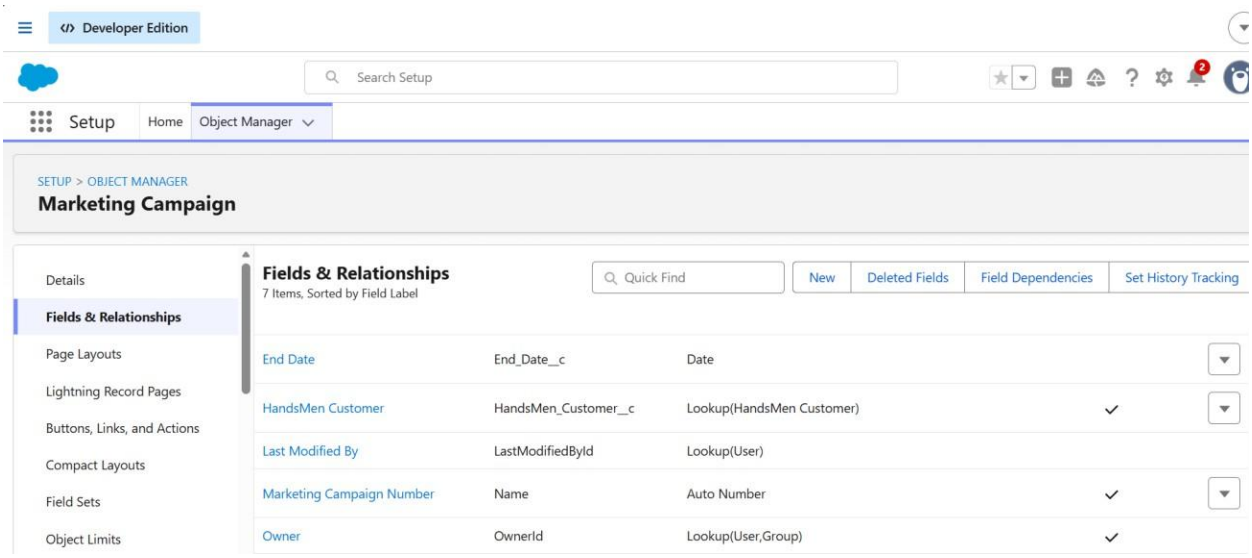


Figure 9. Marketing Campaign fields and customer lookup.

6. Formula Fields and Validation Rules

6.1 Formula Fields

Formula fields calculate and display a value from other fields. The supplied report identifies Stock_Status__c as a Formula (Text) field and mentions a customer Full Name formula. Their exact expressions are not available in the reviewed text, so the expressions must be copied from the live org rather than guessed.

Formula field	Documented status	Required evidence
Inventory.Stock_Status__c	Formula (Text) field is reported.	Formula editor showing the complete expression and threshold, plus records demonstrating each output.
HandsMen Customer Full Name	A Full Name formula is reported in the supplied report.	Formula editor and example output for a customer record.

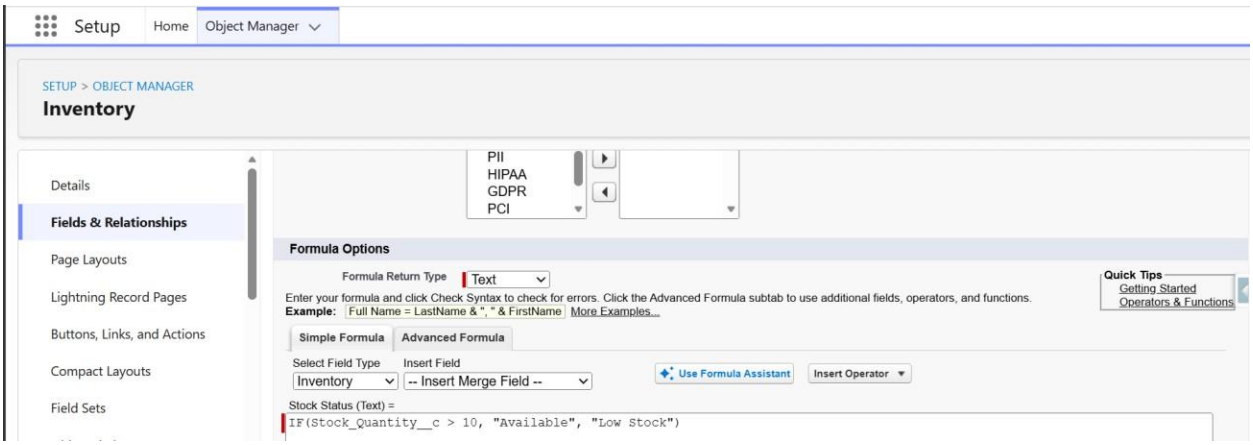


Figure 10. Stock Status formula editor with actual formula visible.

6.2 Validation Rules

The supplied report identifies three active validation rules. The rule names and error messages below are based on that report; the precise formula conditions still need to be checked in Setup.

Object	Rule name	Reported error message/location	Intended control	Evidence status
HandsMen Order	Total_Amount	"Please Enter Correct Amount"; field-level location	Reject amount values that violate the actual configured rule.	Active status reported; formula and save/reject behavior to verify.
Inventory	Stock_Quantity	"the inventory count is never less than zero."; top of page	Prevent negative stock quantity if that is the formula condition.	Active status reported; formula and save/reject behavior to verify.
HandsMen Customer	Email	"Please fill Correct Gmail"; top of page	Enforce the configured email format requirement.	Active status reported; formula and accepted formats to verify.

6.3 Validation Test Cases

Test ID	Test input	Expected result (confirm against actual formula)	Actual result
VAL-01	Save Inventory with negative stock quantity.	Save is blocked and the configured message appears.	Pending — execute and record.
VAL-02	Save Inventory with zero stock quantity.	Should save if zero is permitted by the business rule.	Pending — execute and record.
VAL-03	Save customer with a malformed email.	Save is blocked if it violates the configured email rule.	Pending — execute and record.
VAL-04	Save customer with a valid business-approved email.	Save succeeds if it meets the configured rule.	Pending — execute and record.
VAL-05	Save order with a value below/above the actual allowed range.	Result matches the rule's formula and error message.	Pending — execute and record.

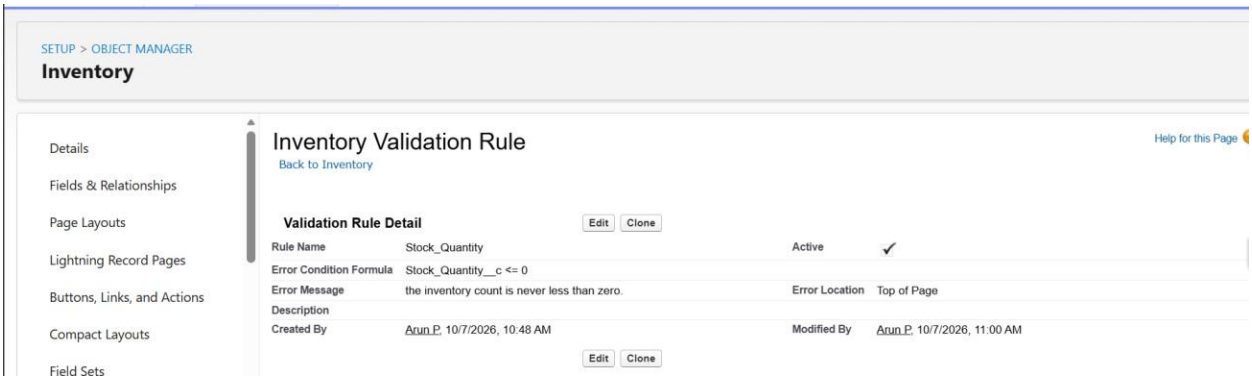


Figure 11. Validation rule details including active checkbox, formula and error message.

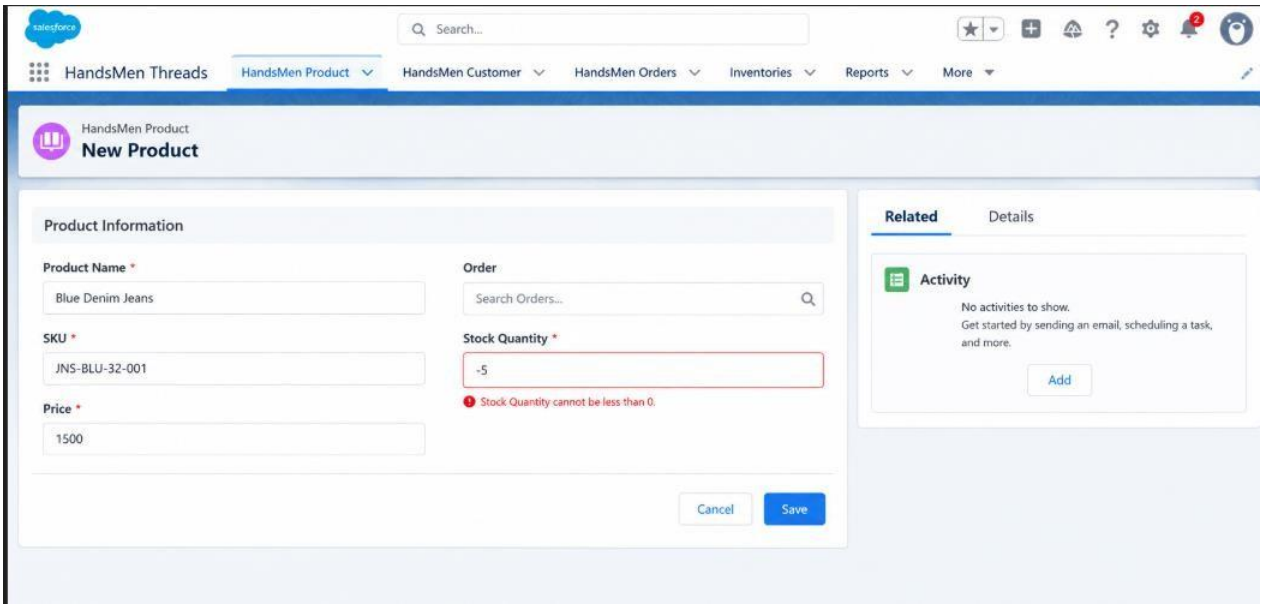


Figure 12. Runtime validation test showing the invalid value and displayed error.

Important: do not describe a rule as preventing negative values or validating Gmail specifically unless its formula actually implements that condition. Use the live formula as the source of truth.

7. Phase 2 — Flow Builder and Email Automation

7.1 Stock Alert Flow

The supplied report describes an active record-triggered flow named Stock Alert Flow V2 on Inventory. It reports a created-or-updated trigger, one condition and a Run Immediately path associated with the Low Stock Alert email alert. The exact threshold, condition logic, recipients and fault behavior must be checked in Flow Builder.

Element	Documented / intended behavior	Verification item
Trigger	Record-triggered flow on Inventory; create/update is reported.	Confirm trigger object, before/after-save setting and entry conditions.
Condition	One condition is reported.	Record the exact field, operator, threshold and condition requirements.
Action	Low Stock Alert email alert is reported.	Confirm recipients, template, record context and email action.
Re-entry / duplicate alerts	Not established.	Test repeated updates and decide whether duplicate notifications are expected.
Fault handling	Not established.	Review fault path and monitoring approach.

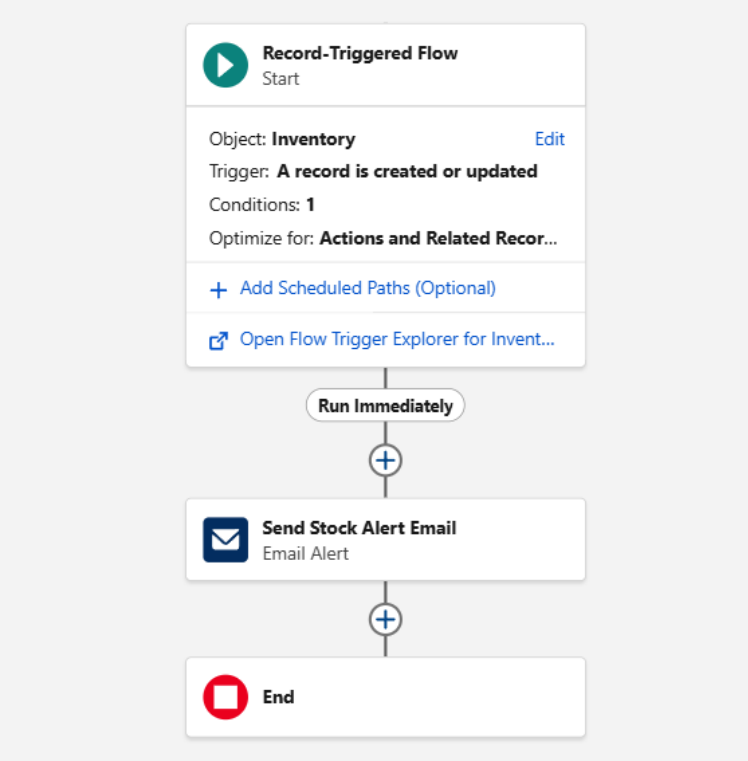


Figure 13. Stock Alert Flow V2 canvas showing start, conditions and email action.

7.2 Loyalty Status Update Flow

The supplied report describes an active scheduled flow named Loyalty Status Update Flow V1. It reports Gold for total purchases above 1000, Bronze below 500, and Silver through a default outcome. Confirm the actual field used, currency/number units, whether boundary values 500 and 1000 are included in a tier, schedule time zone, and how status is updated.

Element	Reported behavior	Verification item
Schedule	Scheduled daily flow is reported.	Confirm frequency, start time, time zone and target object.
Gold outcome	Reported for total purchases above 1000.	Confirm exact condition and boundary.
Bronze outcome	Reported for total purchases below 500.	Confirm exact condition and boundary.
Silver outcome	Reported as default outcome.	Confirm intended range and default behavior.
Email	Loyalty Program Email is associated with loyalty communications in the report.	Confirm whether this flow sends the email and which status changes qualify.

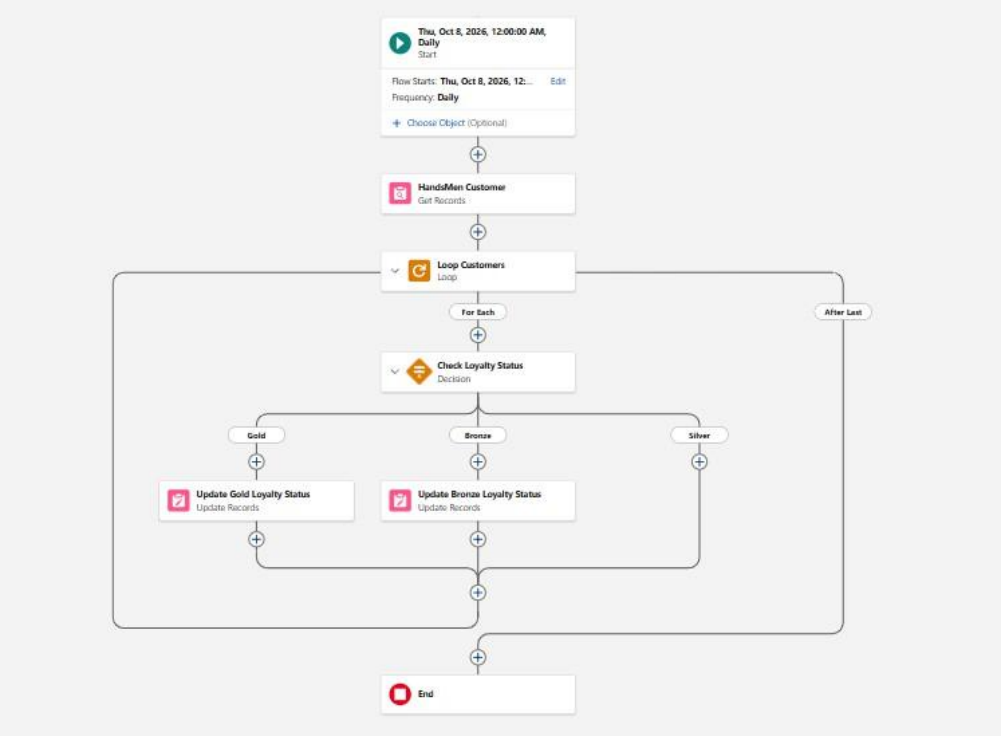


Figure 14. Loyalty Status Update Flow V1 canvas with schedule and decision outcomes.

7.3 Email Templates and Alerts

Template / alert	Reported details	Before claiming success
Low Stock Alert	Classic HTML template marked Available for Use; subject reported as “URGENT: Low Stock Alert”.	Verify full body, merge fields, sender, recipients, alert association and test delivery.
Loyalty Program Email	Classic HTML template marked Available for Use; subject reported as “Welcome to the HandsMen Threads VIP Loyalty Program!”.	Verify full body, merge fields, sender, recipients, consent and test delivery.

A template marked Available for Use is not proof that a message was delivered. Capture a controlled test to an approved mailbox and redact private information before placing evidence in the report.

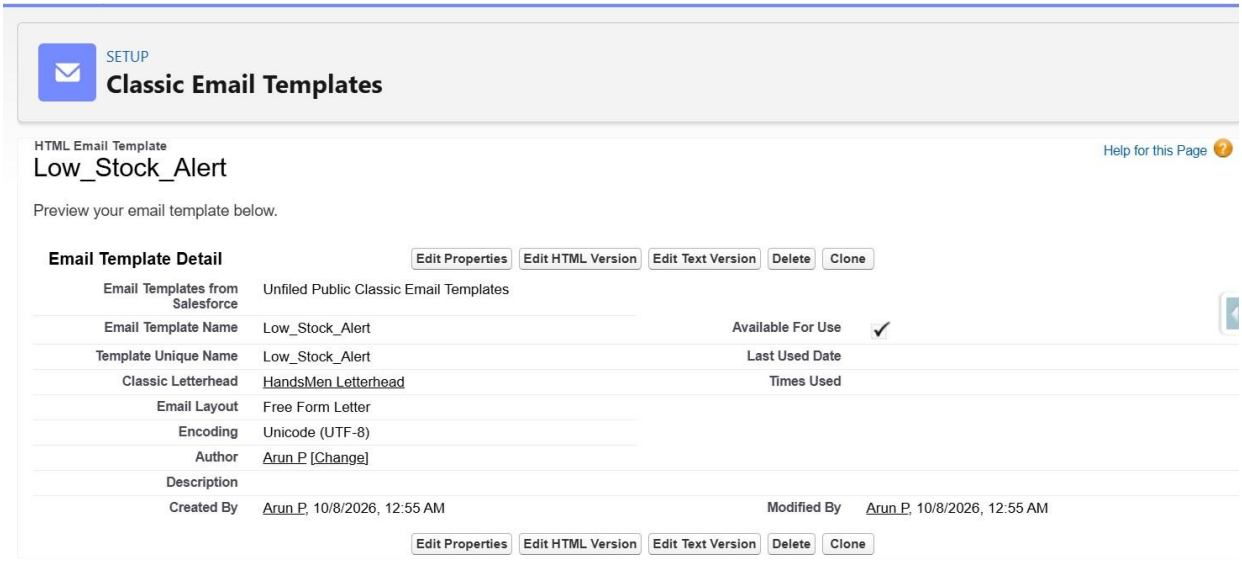


Figure 15. Low Stock Alert email template and email-alert configuration.

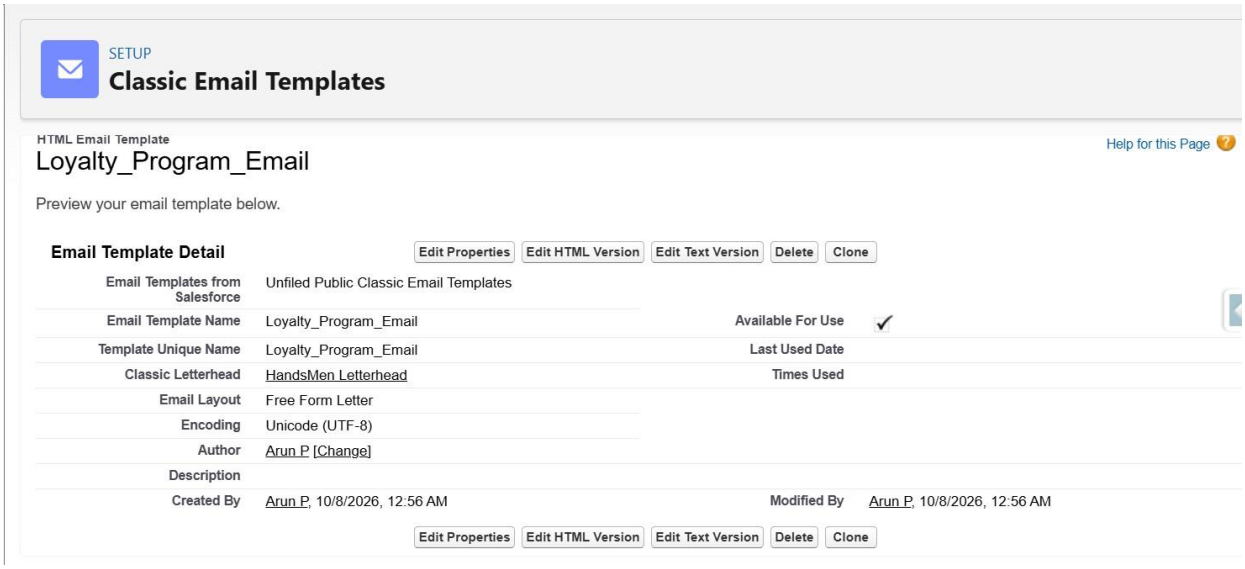


Figure 16. Loyalty Program Email template and email-alert configuration.

7.4 Scheduled Jobs and Asynchronous Processing

The supplied report refers to Scheduled Jobs entries named Daily Inventory Sync and Loyalty_Status_Update_Flow-1, plus a scheduled customer-status update entry. It describes one entry as Scheduled Apex and another as Scheduled Flow. A Scheduled Apex entry does not by itself establish that a Batch Apex class exists or that the job completed successfully.

Item	Current evidence description	Required details
Daily Inventory Sync	Reported in Scheduled Jobs as Scheduled Apex.	Capture Apex class name, schedule, owner, next/last run and job status; attach source code if this is your implementation.
Loyalty_Status_Update_Flow-1	Reported as a Scheduled Flow.	Capture active flow version, schedule, target records and run result.
Scheduled customer-status update	A related scheduled entry is mentioned.	Confirm whether this is a separate job or the same process before describing it.

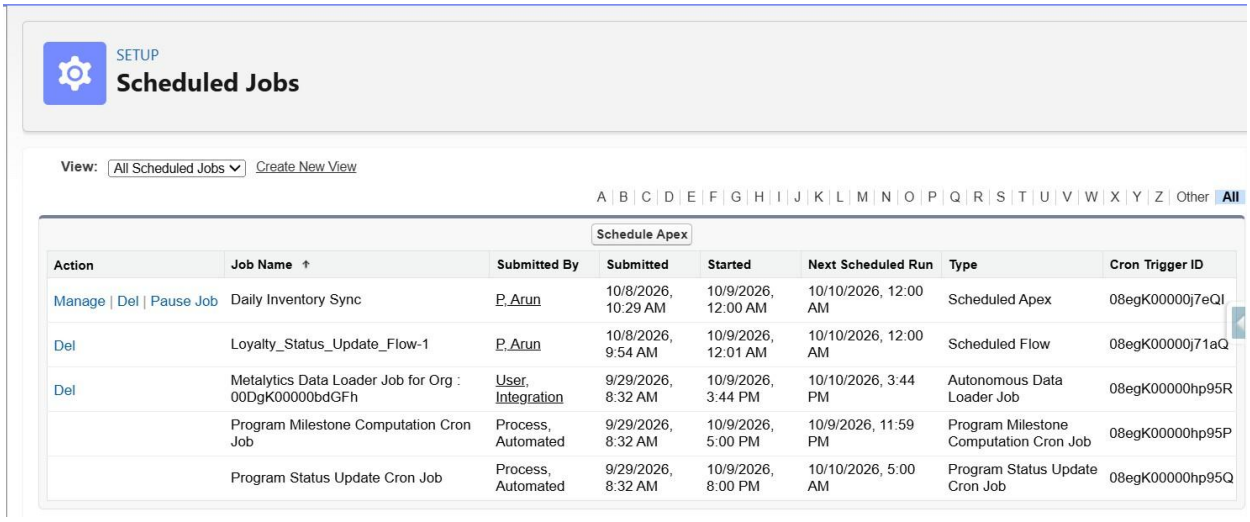


Figure 17. Scheduled Jobs screen showing actual job names, types, schedule and status.

8. Apex Development

8.1 Apex Trigger

Order Apex Trigger: The Order Apex Trigger invokes the Order Trigger Handler class during order record creation or updates. It automates order-related business logic and helps maintain data consistency in the HandsMen Threads Salesforce application.

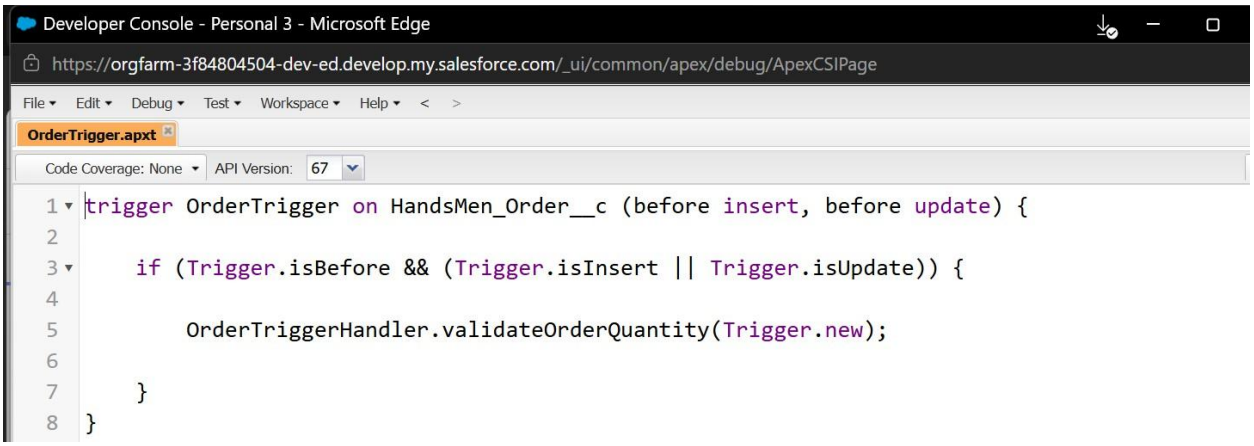


Figure 18. Apex trigger source code in Developer Console..

8.2 Apex Class

Order Trigger Handler Class: The Order Trigger Handler class manages order-related business logic in the HandsMen Threads Salesforce application. It processes order records during events such as creation or updating and helps maintain data consistency through validation and automatic field updates. Separating the logic into a handler class makes the Apex trigger easier to maintain andtest.

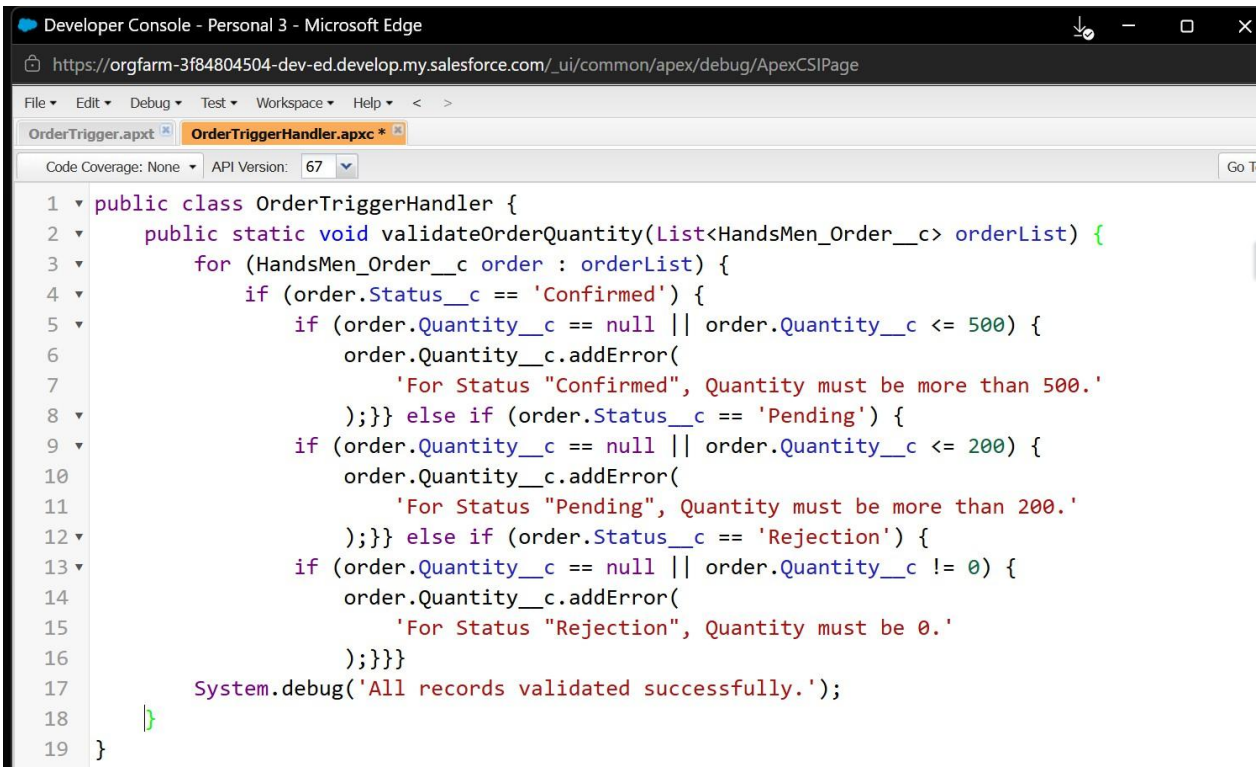


Figure 19. Apex class source code.

8.3 Batch Apex / Scheduled Apex

The InventoryBatchJob class automates inventory processing in Salesforce using the Database.Batchable and Schedulable interfaces. It identifies products with stock quantities below 10 and increases their stock by 50 units. The class supports batch processing, scheduled execution, and error logging to assist inventory management in the HandsMen Threads application.

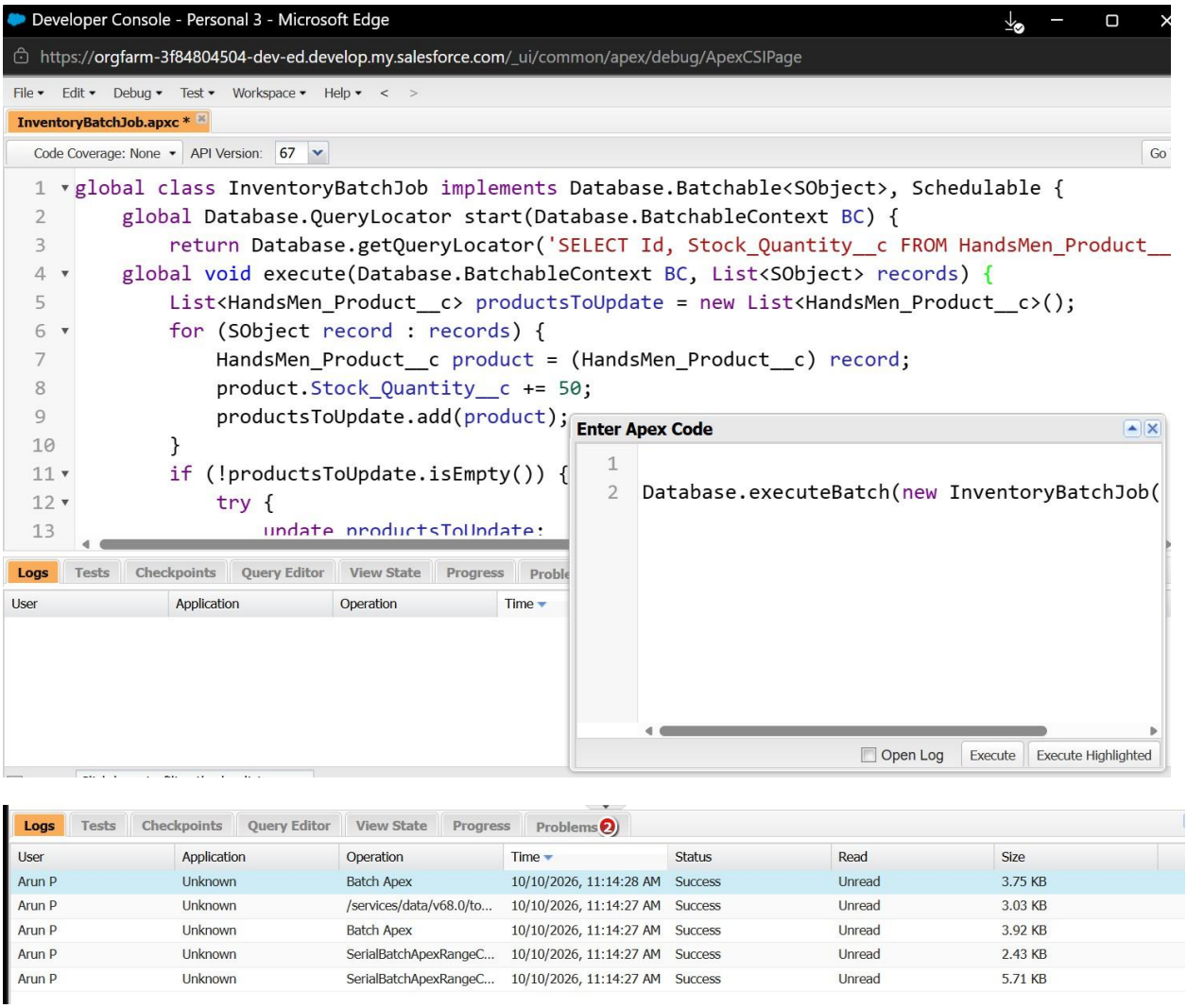


Figure 20. Batch/Scheduled Apex class and successful Apex test result..

8.4 Apex Test Classes

If Apex is implemented, test classes should cover normal behavior, boundary values, error paths and bulk operations. Record actual test method names and the test-run result. Never invent coverage percentages or mark tests as passed without a Salesforce test result.

Test scenario	Expected outcome	Actual outcome
Valid input / standard path	Business operation completes correctly.	Pending execution / enter result.
Boundary values	Rules and calculations behave at thresholds.	Pending execution / enter result.
Bulk records	Multiple records process without governor-limit errors.	Pending execution / enter result.
Invalid input / error path	Invalid operation is handled as designed.	Pending execution / enter result.

9. Phase 3 — UI/UX and Lightning App

9.1 Lightning App Setup

The HandsMen Threads app should provide clear navigation to the custom objects most frequently used by each team. Confirm the app name, navigation items, branding and assigned profiles in App Manager. Do not state that a page layout or Lightning record page has been customized unless the relevant setup has been inspected.

9.2 Tabs, Page Layouts and Lightning Pages

- Capture the custom tab definitions for HandsMen Customer, HandsMen Product, HandsMen Order, Inventory and Marketing Campaign, where configured.
- Review page layouts so key fields appear in a logical order and required fields are clear.
- If Dynamic Forms or custom Lightning record pages are used, document the actual field sections, visibility rules and activation assignments.
- Ensure the navigation labels are consistent and correct spelling errors such as “Inventorys” if that is the actual tab label.

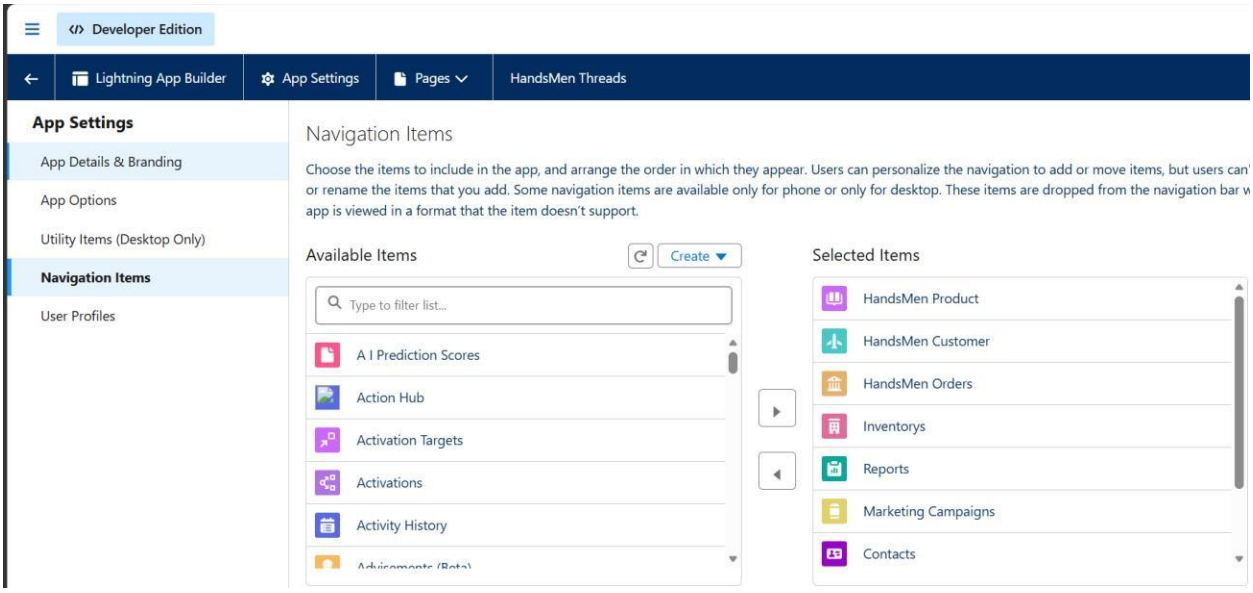


Figure 21. App Manager configuration and app navigation items.

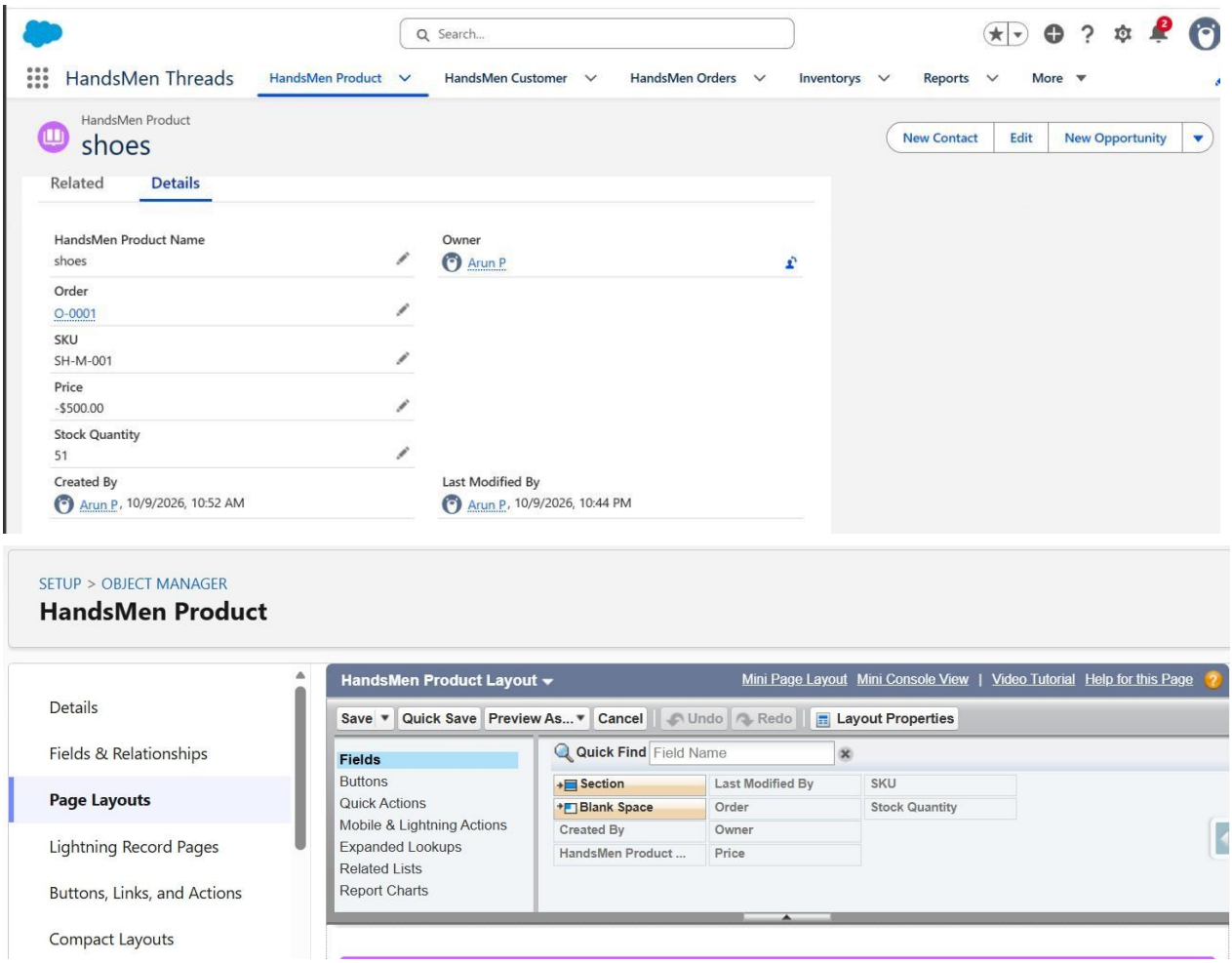


Figure 22. A representative record page / page layout with key fields.

10. Reports and Dashboards

Reports and dashboards are required only if they have been created in the org. If available, document the report type, filters, grouping, key metrics, folder access and dashboard components. If not implemented, list them as a proposed enhancement rather than claiming they exist.

Suggested report	Business question	Status
Low Stock by Warehouse	Which inventory records need replenishment?	Proposed unless confirmed in the org.
Orders by Status	How many orders are open, completed or cancelled?	Proposed unless confirmed in the org.
Customer Loyalty Summary	How are customers distributed by loyalty tier?	Proposed unless confirmed in the org.
Campaign Customer List	Which customers are associated with each campaign?	Proposed unless confirmed in the org.

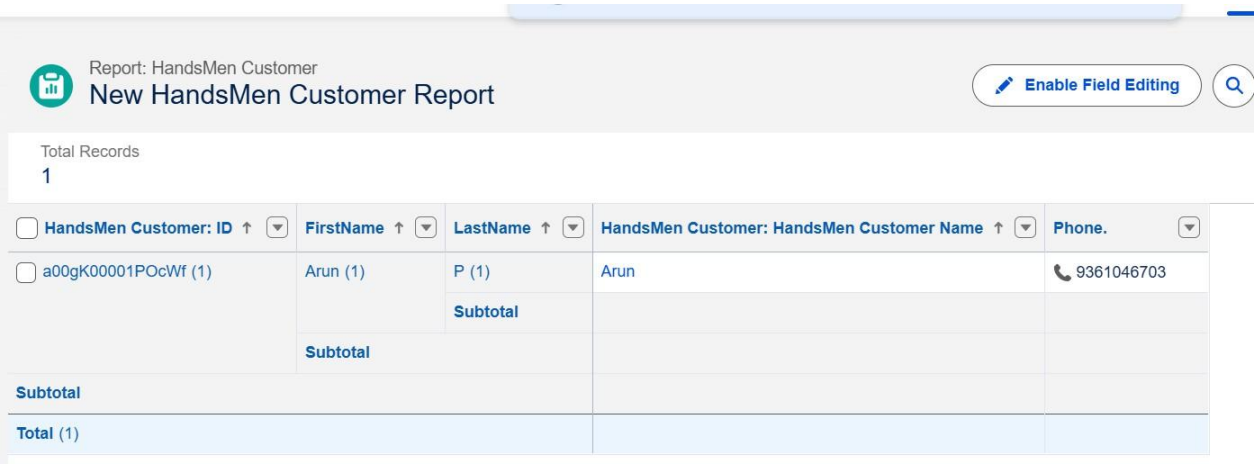


Figure 23. HandsMen report

11. Phase 4 — Security and User Management

11.1 Profiles and Permission Sets

Profiles provide baseline user permissions; permission sets can grant additional permissions. The supplied report identifies a custom profile named Platform 1 and reports zero entries in its visible Enabled Apex Class Access and Enabled Flow Access sections at the time of capture. This does not prove the permissions of every user or profile. Verify object permissions, field-level security, Apex class access and flow access against actual project users.

11.2 Roles and Role Hierarchy

The supplied report describes Sales, Inventory and Marketing roles under KEC SOLUTIONS > CEO, with no users assigned in the screenshots. Role hierarchy primarily affects record visibility in conjunction with sharing settings; it does not itself grant object or field permissions.

Role / persona	Expected responsibility	Permission review
Sales	Customer and order work.	Customer/order object access; appropriate contact and order fields.
Inventory	Inventory quantity, warehouse and replenishment work.	Inventory/product access; restrict unnecessary customer data.
Marketing	Campaign and approved loyalty communication work.	Campaign/customer access; review contact-data visibility.
Administrator	Configuration, deployment and monitoring.	Elevated access only for authorized administrators.

11.3 Sharing and Field-Level Security

Before submission, record the actual organization-wide defaults, sharing rules, permission sets and field-level security. If no sharing rules or permission sets have been created, state that clearly and identify the intended future configuration.

11.4 Field History and Duplicate Management

The project guidelines request field history tracking and duplicate/matching rules to be documented. Confirm whether these are configured. If enabled, list the tracked fields, matching rule, duplicate rule action and user-facing message. If not enabled, mark them as not implemented or proposed.

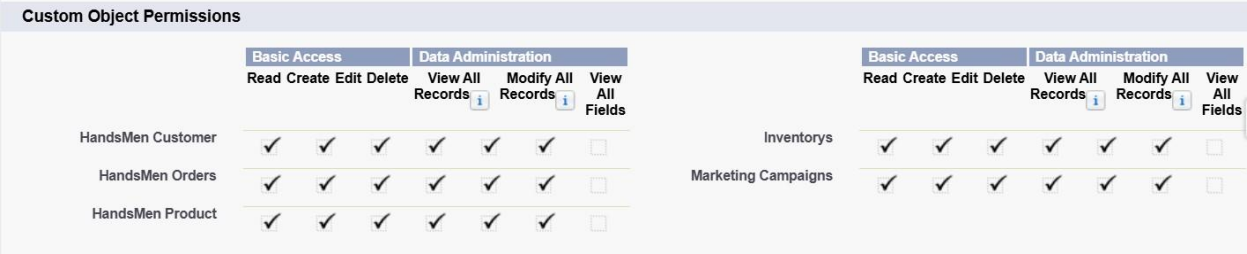


Figure 24. permission-set object and field permissions.

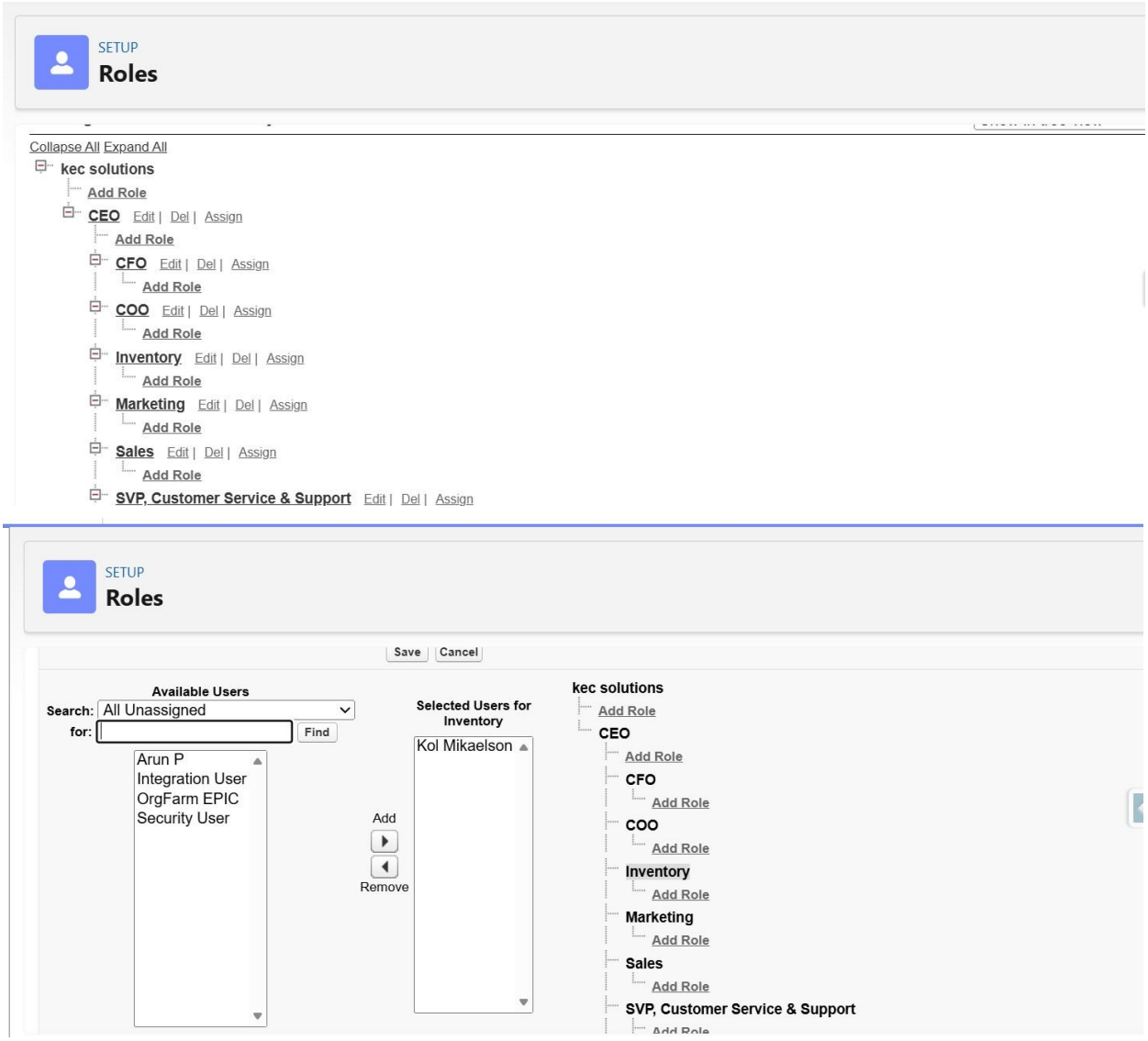


Figure 25. Role hierarchy and role assignment evidence.

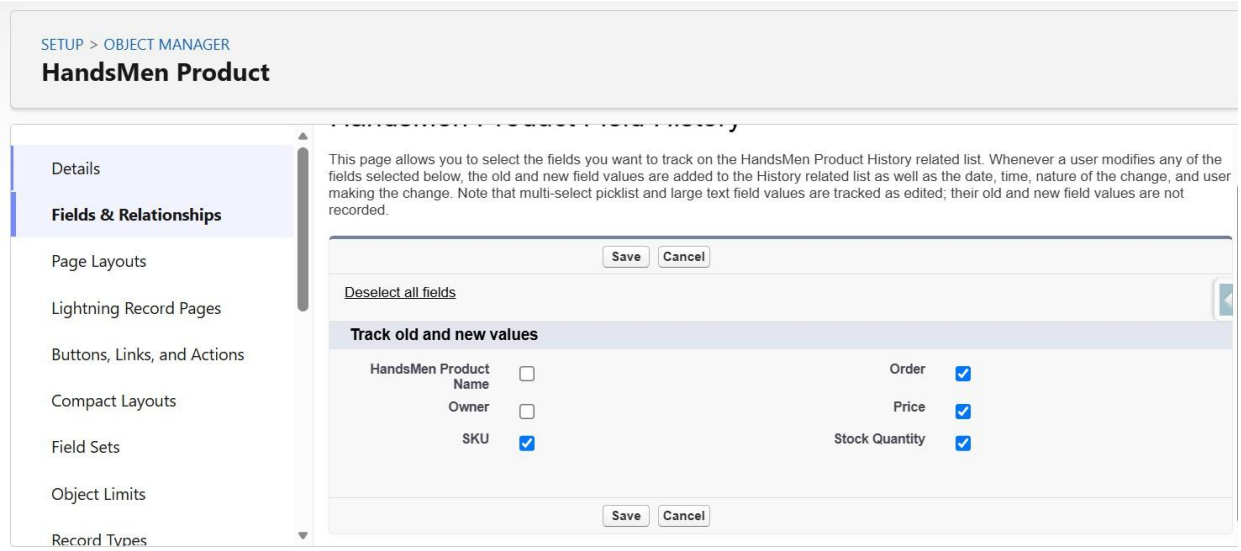


Figure 26. Sharing, duplicate rules or field history configuration.

12. Data Loading and Migration

Document the data-loading method actually used, such as manual record entry, Data Import Wizard or Data Loader. For a repeatable import, prepare a field-mapping sheet, validate lookup keys, remove duplicates, test a small sample and preserve the import success/error results.

Step	Recommended control	Evidence to retain
Prepare data	Remove duplicates, standardize formats and validate required fields.	Sanitized sample file and field mapping.
Map fields	Match CSV columns to Salesforce fields and lookup identifiers.	Mapping screenshot or import configuration.
Test import	Load a small sample first.	Sample record screenshots and error report.
Validate	Check record counts, relationships and validation outcomes.	Post-import reconciliation.
Recover errors	Review failed rows and correct only intended values.	Error file and corrective action notes.

If no data migration was performed, state that records were entered manually or that migration was not part of this version. Do not claim use of Data Loader without evidence.

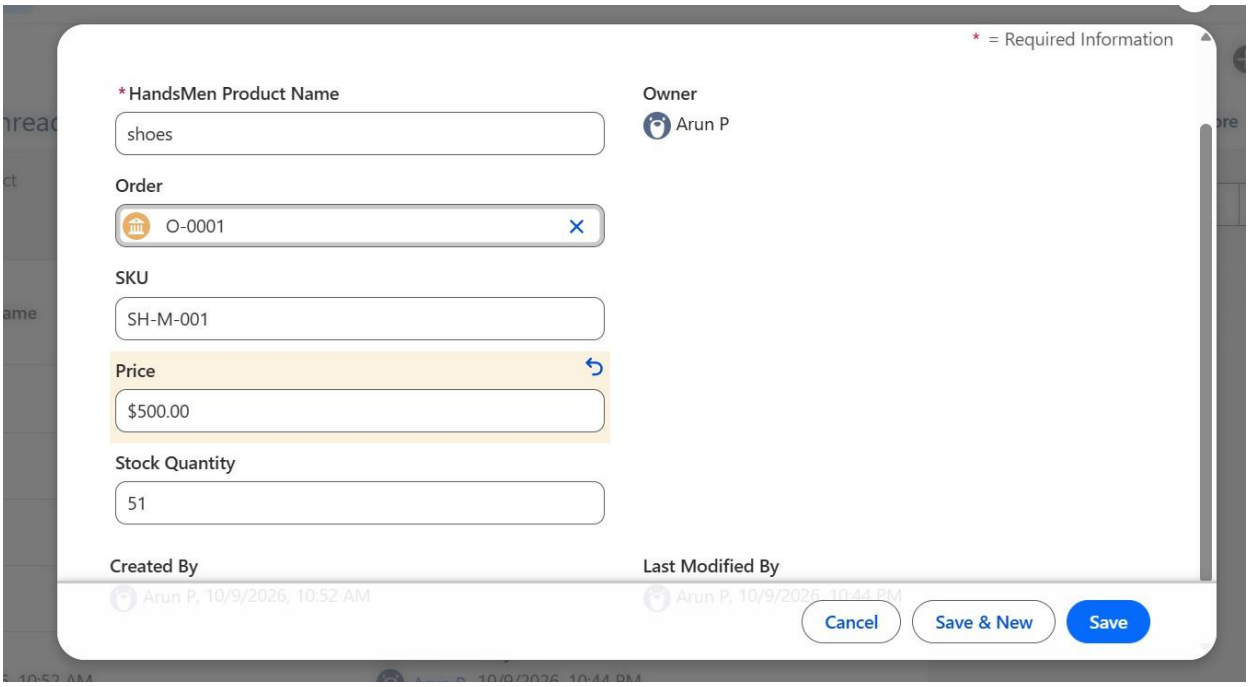


Figure 27. sample record entry.

13. Phase 4 — Testing Strategy and Results

13.1 Testing Approach

Testing should verify configuration and behavior separately. Configuration tests inspect metadata; functional tests create or update records and compare actual outcomes with expected outcomes; security tests use representative users; and automation tests inspect flow debug output, job history and email delivery. Use sanitized test records and avoid exposing real customer information.

13.2 Functional Test Matrix

ID	Feature	Test input/action	Expected result	Actual result / evidence
TC-01	Customer creation	Create a customer with valid required values.	Record saves and fields display correctly.	Pending — enter actual result and figure number.
TC-02	Email validation	Use an invalid email value.	Save is blocked if the value violates the configured formula.	Pending.
TC-03	Order creation	Create an order linked to an existing customer.	Order saves and relationship opens correctly.	Pending.
TC-04	Order amount rule	Enter a value that violates the actual Total_Amount rule.	Save is blocked and configured message appears.	Pending.
TC-05	Negative inventory	Set stock quantity below zero.	Save is blocked if the actual rule prevents negative stock.	Pending.
TC-06	Stock Status formula	Use records on both sides of each configured threshold.	Formula output matches the actual expression.	Pending.
TC-07	Stock Alert Flow	Create/update Inventory to meet the configured condition.	Configured alert action runs once as designed.	Pending; attach Flow/debug/email evidence.
TC-08	Loyalty Flow	Set customer data to each loyalty boundary/range.	Correct tier is assigned according to actual criteria.	Pending.
TC-09	Email template	Send a test email to an approved test mailbox.	Correct subject/body/merge fields are received.	Pending.
TC-10	Role/security	Log in as Sales, Inventory and Marketing test users.	Each user sees only permitted records and fields.	Pending.
TC-11	Scheduled processing	Inspect run history after scheduled execution.	Job/flow completes for intended records without errors.	Pending.
TC-12	Apex tests	Run actual Apex test class if Apex exists.	Tests pass and cover key branches/bulk cases.	Not applicable until Apex implementation is confirmed.

13.3 Test Evidence Standard

- For each test, record test ID, date, test user, input values, expected result and actual result.
- Attach screenshots of both the input/action and the resulting output where applicable.
- For failed tests, record the defect, severity, correction and retest result.
- For flows, retain debug details or execution evidence; for scheduled jobs, retain job status/history; for email, retain a safe delivery confirmation.
- Never fill the Actual Result column with “Pass” until the test has been executed.

Inventory Number

I-0001

Product

Black T shirt

Stock Quantity

3

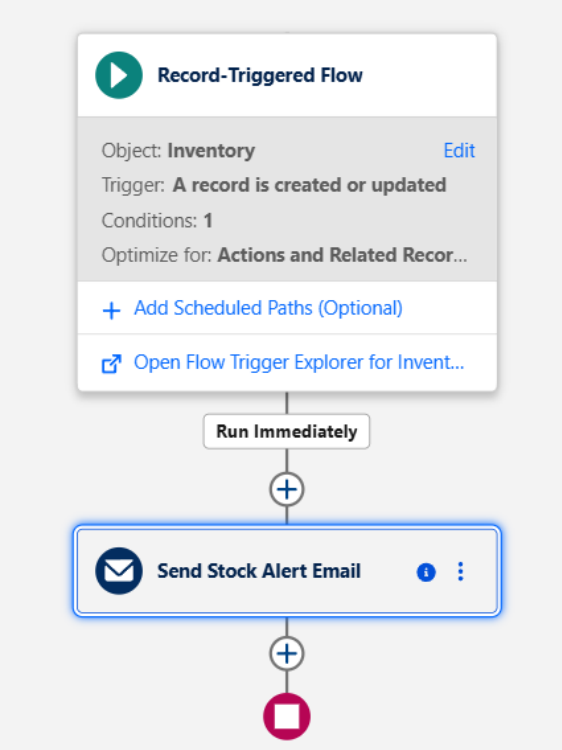
Stock Status

Low Stock

This field is calculated upon save

Warehouse

Coimbatore storage



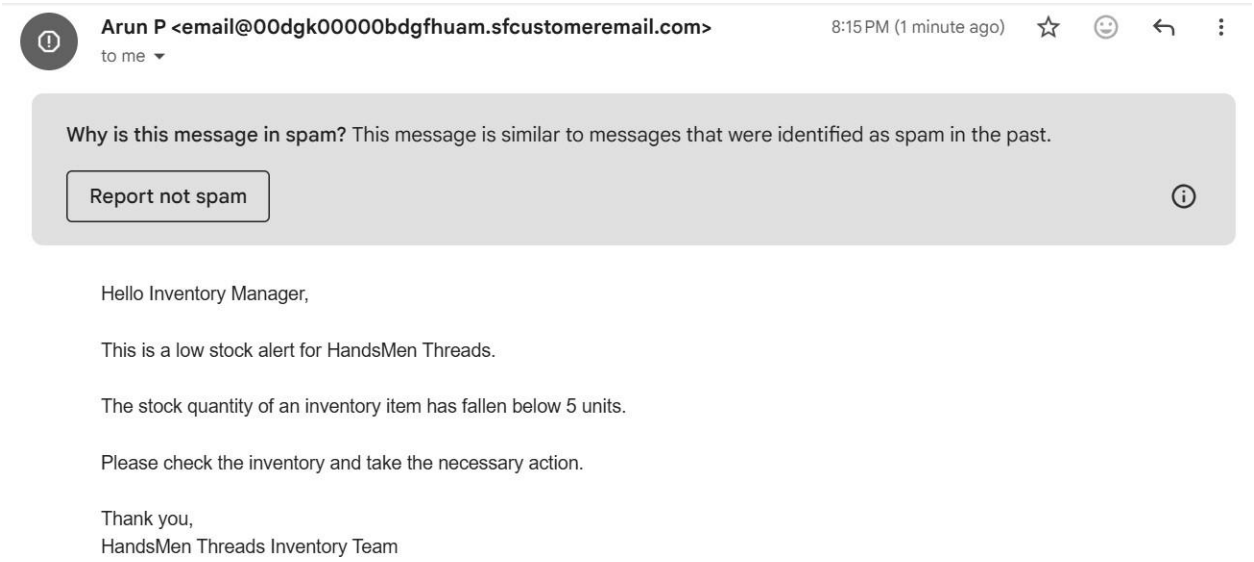


Figure 29. Stock Flow email delivery

14. Error Handling and Troubleshooting

Symptom	Checks	Evidence / corrective action
Validation blocks an expected valid record	Inspect the exact formula, field type, blank handling and input format.	Capture error and formula; correct only after confirming business requirements.
Stock alert does not send	Check flow activation/version, entry conditions, recipient resolution, email alert, deliverability and fault path.	Review Flow debug, error email/log and test mailbox.
Loyalty tier is incorrect	Check source field, threshold boundaries, schedule, default outcome and whether total purchases is current.	Test below, at and above every boundary.
Scheduled job does not run	Check job status, owner, time zone, schedule and Apex/flow errors.	Retain Scheduled Jobs / job detail evidence.
User cannot access a record or field	Check profile, permission sets, object permissions, field-level security, role and sharing.	Re-test as that user after changes.
Import fails	Check field mappings, required fields, duplicate constraints, lookup IDs and validation rules.	Save the error file and document corrected rows.

Use Salesforce Setup, Flow Builder debug, Apex test results and available debug logs to investigate issues. Record the root cause and corrective action rather than only noting that the issue was fixed.

15. Phase 5 — Deployment Strategy

Choose a deployment method supported by the project environment. For a small academic project, change sets may be appropriate when source and target orgs support them; Salesforce CLI and a version-controlled metadata workflow are preferable when available. The report should identify the method actually used.

1. Inventory the metadata to move: objects, fields, relationships, validation rules, formulas, flows, email templates, Apex, tabs, Lightning pages and permission components.
2. Check dependencies and API names; ensure referenced fields, templates, classes and recipients exist in the target org.
3. Deploy to a test/sandbox org when available and run the test matrix.
4. Deploy to the target org during an agreed window, then activate only the intended flow versions and schedules.
5. Perform smoke tests for record creation, validation, stock alerts, loyalty updates, email delivery and access.
6. Retain deployment notes, test results, known limitations and a rollback/recovery plan.

16. Maintenance and Monitoring

- Review failed flow interviews, scheduled job status and Apex exceptions at an agreed cadence.
- Re-test validation rules and formulas when fields or business thresholds change.
- Review email recipients, sender identity, template wording and deliverability settings.
- Audit user assignments, profiles, permission sets, sharing rules and field-level security.
- Maintain a change log and update screenshots when configuration changes.
- Periodically verify backups/export availability and follow the organization's data retention requirements.
- Document issue owner, date detected, severity, root cause, fix and retest.

16.1 Operational Ownership

Activity	Suggested owner	Frequency
Flow and scheduled job monitoring	Salesforce administrator / project owner	Weekly or after changes
User access review	Administrator / project supervisor	Monthly or on role changes
Data quality review	Business owner for each object	Monthly
Email template review	Marketing / inventory owner	Before campaigns and after changes
Regression testing	Developer and assigned testers	Every release

17. Planning and Development Milestones

Milestone	Work	Completion evidence
1. Requirements	Define users, use cases, scope and acceptance criteria.	Approved requirements and test matrix.
2. Data model	Create objects, fields and relationships.	Object Manager screenshots and relationship review.
3. Data quality	Configure formulas and validation rules.	Formula/rule screenshots and positive/negative tests.
4. Automation	Configure and activate applicable flows and scheduled processes.	Flow canvas, active version, debug and run history.
5. Communication	Configure templates and alerts.	Template preview, alert setup and delivery test.
6. Security/UI	Configure app, tabs, layouts, roles and permissions.	App navigation and user-based access tests.
7. Verification	Run functional, bulk and failure-path tests.	Completed test matrix, logs and issue list.
8. Deployment	Deploy reviewed metadata and run smoke tests.	Deployment notes and post-deployment results.

18. Future Enhancements

- Introduce an Order Item junction object for multi-product orders and line-level pricing.
- Establish one authoritative inventory quantity source and define stock reduction/replenishment behavior.
- Make low-stock thresholds configurable and document alert frequency/duplicate-alert policy.
- Formalize loyalty tiers, points, threshold boundaries and status history.
- Add campaign response, conversion and revenue attribution fields if required by the business.
- Create reports and dashboards for orders, low stock, loyalty distribution and campaign outcomes.
- Add duplicate/matching rules and field history tracking if appropriate for the data and privacy requirements.
- If needed, develop bulk-safe Apex with test classes and monitored scheduled/batch execution.
- Explore chatbot or AI-based product/customer suggestions only after data quality, access controls and business requirements are established.

19. Conclusion

HandsMen Threads provides a structured Salesforce CRM design for customer, product, order, inventory and marketing-campaign records. The reviewed project material describes custom objects and relationships, active validation rules, formula fields, Flow Builder automation, email templates, scheduled processing entries and role configuration.

The main improvement in this report is the separation of verified configuration from intended behavior and unexecuted tests. Before final submission, the author should insert screenshots from the actual org, confirm all field/API names and formulas, document the exact flow conditions and schedule, and complete the functional and security test matrix. Apex Trigger, Apex Class and Batch Apex should be reported as completed only if their source code and test results exist.

Once screenshots and actual test outcomes are added, the report can serve as a traceable record of the implementation, its limitations and the steps required for maintenance